

Charles Hart

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EDUCATION

Cornell University, Duffield College of Engineering, Ithaca, NY
Master of Engineering, Systems Engineering, **GPA: 4.14**

Expected December 2026

Cornell University, Duffield College of Engineering, Ithaca, NY
Bachelor of Science, Mechanical Engineering, **GPA: 3.43**
Dean's List: Fall 2024, Spring 2025, Fall 2025, Spring 2026

May 2026

Relevant Courses: Systems Integration, Verification, & Validation; Optimization Control & Decision Theory; Fluid Mechanics; System Dynamics; Mechanics of Materials; Mechatronics; Thermodynamics; Heat Transfer; Automotive Engineering; Electric Vehicles; Uncertainty Analysis for Engineers (Statistical Data Analysis using Python)

SPECIALIZED SKILLS

Technical: Autodesk Inventor, MATLAB, LaTeX, Data Analysis with Python/Excel, C++, ANSYS, Shop Machining/Drawings

Professional: Stakeholder Engagement, Empathy Mapping, Multitasking, Customer Service, Problem Solving, Strategic Thinking, Teamwork, Group Dynamics, Quick Learning, Organization

Languages: English (Native); Spanish (Professional Working Proficiency); Italian (Elementary Proficiency)

ENGINEERING EXPERIENCE

Cornell University Sustainable Design, Cornell University, *City/Mobility Team Member*

Jan 2026 - Present

- Conducted empathy-based field research in the Ithaca Commons, observing user behavior and engaging with visitors to identify barriers preventing Cornell students from visiting the downtown district
- Collaborated on multidisciplinary team to develop community-focused design concepts aimed at improving accessibility, transportation connections, and student engagement with local businesses
- Generated and evaluated urban mobility and public space concepts including transit information improvements, bus stop enhancements, and micromobility infrastructure
- Synthesized field observations and stakeholder insights into actionable design proposals presented to project partners and local community stakeholders

Cornell Mars Rover Project Team, Cornell University, *Drives Sub-Team Member*

Oct 2022 - Jan 2025

- Designed and produced a lightweight wheel-hub assembly utilizing 3D-printed tires and aluminum hubs, reducing weight from previous rover iterations for annual competition
- Identified external requirements set for the rover system by competition rules and established internal parameters to ensure compatibility with other sub-systems and overall rover system
- Became familiar with material sourcing processes from external vendors for rover component fabrication.
- Modeled components and built sub-system assemblies using CAD software (Autodesk Inventor) and performed stress and failure analysis using ANSYS
- Produced detailed machine-shop drawings and manufactured components using a mill and lathe equipment

ACADEMIC PROJECTS

Modular Sustainable Bus Shelter Capstone Project, *Cornell University Sustainable Design (CUSD) Mobility Team, Howard N. McManus Jr. Design Award Recipient*

Jan-May 2026

- Designed a modular base-to-foundation connection system for a sustainable transit shelter
- Performed structural analyses for wind, snow, ice, and dead-load conditions using ASCE standards
- Developed and refined CAD designs through multiple design iterations to improve structural performance
- Validated design performance using ANSYS FEA and identified critical stress concentrations

OTHER EXPERIENCE

Statler Hotel's Regent Lounge, Ithaca, NY, *Server*

Mar 2023 - Present

- Work with a team of other servers and knowing how to continuously redistribute work evenly in a dynamically demanding environment
- Quickly memorize extensive food and beverage menus, while understanding allergens, liquor lists, and cocktail composition
- Maintain high-quality customer service while paying close attention to details and customers' needs